

Assignment:-1

Task 1:- List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

Three Real-Life Uses of a Multimeter

1. Computer not turning on — A multimeter can check the PSU voltage to see if the power supply is working properly.
2. Laptop battery not charging — A multimeter can check the charger's output voltage to find out if the charger is providing power.
3. Smartphone not charging — A multimeter can check the USB charger or charging port voltage to find a power problem.

Task 2:- Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video. Focus on the codes shown and how

POST Card Video Summary

The POST card shows **numbers or codes** when the computer starts.

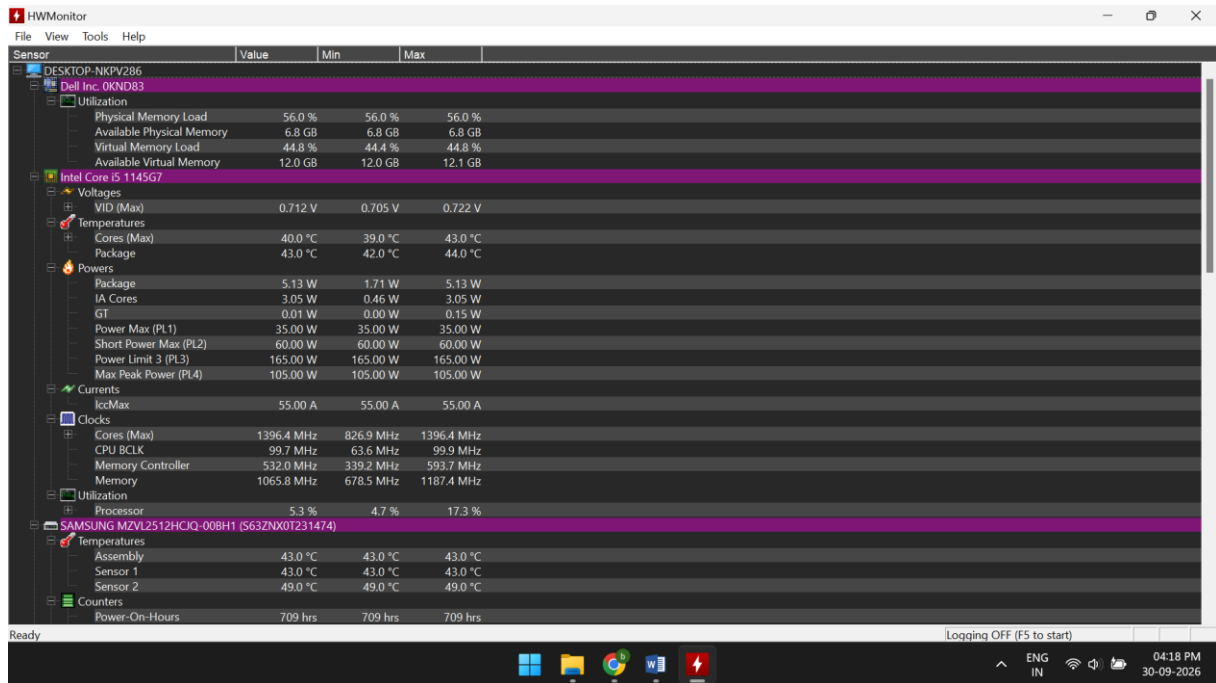
These codes show that the motherboard is checking the **CPU, RAM, and other hardware**.

If the computer stops at a code, it can help find the **hardware problem**.

For example, a code may point to a problem with **RAM, CPU, or graphics hardware**.

<https://youtu.be/e5vqz6zRV6s?si=RD1NvK7GOkRRgz-n>

Task 3:- Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details. Blur or crop out any personal information before submitting.



The screenshot shows the HWMonitor application window. The interface includes a menu bar (File, View, Tools, Help) and a table of system sensors. The sensors are categorized into Utilization, Voltages, Temperatures, Powers, Currents, Clocks, and Counters. The table has columns for Sensor, Value, Min, and Max. The system information at the bottom indicates the system is ready, logging is off, and the date is 30-09-2026.

Sensor	Value	Min	Max
DESKTOP-NKPV286			
Dell Inc. 0KND83			
Utilization			
Physical Memory Load	56.0 %	56.0 %	56.0 %
Available Physical Memory	6.8 GB	6.8 GB	6.8 GB
Virtual Memory Load	44.8 %	44.4 %	44.8 %
Available Virtual Memory	12.0 GB	12.0 GB	12.1 GB
Intel Core i5 1145G7			
Voltages			
VID (Max)	0.712 V	0.705 V	0.722 V
Temperatures			
Cores (Max)	40.0 °C	39.0 °C	43.0 °C
Package	43.0 °C	42.0 °C	44.0 °C
Powers			
Package	5.13 W	1.71 W	5.13 W
IA Cores	3.05 W	0.46 W	3.05 W
GT	0.01 W	0.00 W	0.15 W
Power Max (PL1)	35.00 W	35.00 W	35.00 W
Short Power Max (PL2)	60.00 W	60.00 W	60.00 W
Power Limit 3 (PL3)	165.00 W	165.00 W	165.00 W
Max Peak Power (PL4)	105.00 W	105.00 W	105.00 W
Currents			
IccMax	55.00 A	55.00 A	55.00 A
Clocks			
Cores (Max)	1396.4 MHz	826.9 MHz	1396.4 MHz
CPU BCLK	99.7 MHz	63.6 MHz	99.9 MHz
Memory Controller	532.0 MHz	339.2 MHz	593.7 MHz
Memory	1065.8 MHz	678.5 MHz	1187.4 MHz
Utilization			
Processor	5.3 %	4.7 %	17.3 %
SAMSUNG MZVL2512HCJQ-008H1 (S63ZNX0T231474)			
Temperatures			
Assembly	43.0 °C	43.0 °C	43.0 °C
Sensor 1	43.0 °C	43.0 °C	43.0 °C
Sensor 2	49.0 °C	49.0 °C	49.0 °C
Counters			
Power-On-Hours	709 hrs	709 hrs	709 hrs

Task 4:- Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it checks	Example Problem Detected
Multimeter	Checks voltage, current, and electrical connections.	Faulty power supply (PSU)
POST Card	Checks startup/POST codes from the motherboard.	RAM or CPU problem
Software Diagnostics	Checks hardware health and temperatures using software.	Overheating CPU or failing hard drive